

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

First Semester of BSc Physics Examination March 2018

PD105 Three Dimensional Geometry

Date: 28/03/2018, Wednesday

Time: 10:00 a.m. to 10:30 a.m.

Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer/s and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

1. The distance between the two points $\left(-\frac{C}{A}, 0\right)$ and $\left(0, -\frac{C}{B}\right)$ when axes are rectangular is _____.
(i) $\sqrt{A^2 + B^2}$ (iii) $\frac{1}{AB}\sqrt{A^2 + B^2}$
(ii) $\frac{C}{AB}\sqrt{A^2 + B^2}$ (iv) $C\sqrt{A^2 + B^2}$
2. The polar coordinate form of equation of $x^2 + y^2 = ay$ is _____.
(i) $r = a \sin \theta$ (iii) $r = \sin \theta$
(ii) $r = a \cos \theta$ (iv) $r = \cos \theta$
3. Consider the points $A(-5, -3)$ and $B(5, 2)$. A point $C(-1, -1)$ divides the line joining the points A and B . Then $AC : CB =$ _____.
(i) 3:2 (iii) -2:3
(ii) -3:2 (iv) 2:3
4. The equation to the straight line which cuts off intercepts $a = 3$ with x -axis and $b = 4$ with y -axis is _____.

- (i) $3x + 4y - 12 = 0$ (iii) $4x + 3y - 12 = 0$
(ii) $3x - 4y - 12 = 0$ (iv) $4x - 3y - 12 = 0$
5. The angle between the pairs of straight lines $x - y\sqrt{3} = 5$ and $\sqrt{3}x + y = 7$ is _____.
- (i) 90° (iii) 45°
(ii) 30° (iv) 60°
6. If the coordinates of vertexes of triangle are $(5, 2)$, $(-9, -3)$ and $(-3, -5)$ respectively. Then area of triangle is _____.
- (i) 59 (iii) 53
(ii) 29 (iv) 35
7. If the two straight lines $3x + 2ky - 2 = 0$ and $2x + 5y + 1 = 0$ are parallel, then the value of k is _____.
- (i) $-\frac{5}{4}$ (iii) $\frac{2}{5}$
(ii) $\frac{15}{4}$ (iv) $\frac{3}{2}$
8. The zeros of the quadratic polynomials $ax^2 + bx + c$, $c \neq 0$ are equal then _____.
- (i) c and a have same signs (iii) c and b have opposite signs
(ii) c and a have opposite signs (iv) c and b have same signs
9. If the radius of circle $x^2 + y^2 - 4x + fy - 41 = 0$ is $\sqrt{61}$ then $f =$ _____.
- (i) 8 (iii) -8
(ii) 4 (iv) -4
10. The straight line $y = mx + c$ is tangent to a circle $x^2 + y^2 = a^2$ if _____.
- (i) $c^2 < a^2(1 + m^2)$ (iii) $c^2 < a^2/(1 + m^2)$
(ii) $c^2 > a^2(1 + m^2)$ (iv) $c^2 = a^2(1 + m^2)$
11. The value of $\sin 75^\circ =$ _____.
- (i) $\frac{\sqrt{3} + 1}{2\sqrt{2}}$ (iii) $\frac{\sqrt{3} + 1}{\sqrt{2}}$
(ii) $\frac{\sqrt{3} - 1}{2\sqrt{2}}$ (iv) $\frac{\sqrt{3} - 1}{\sqrt{2}}$
12. The values of $\sin 22\frac{1}{2}^\circ$ and $\cos 22\frac{1}{2}^\circ$ are _____ and _____ respectively.
- (i) $\sqrt{2 - \sqrt{2}}, \sqrt{2 + \sqrt{2}}$ (iii) $\sqrt{2 + \sqrt{2}}, \sqrt{2 - \sqrt{2}}$
(ii) $\frac{1}{2}\sqrt{2 + \sqrt{2}}, \frac{1}{2}\sqrt{2 - \sqrt{2}}$ (iv) $\frac{1}{2}\sqrt{2 - \sqrt{2}}, \frac{1}{2}\sqrt{2 + \sqrt{2}}$
13. The equation to the plane through $(1, 2, 3)$ parallel to $3x + 4y - 5z = 0$ is _____

- (i) $4x + 3y - 5z + 5 = 0$ (iii) $3x - 4y + 5z - 10 = 0$
(ii) $3x + 4y - 5z + 4 = 0$ (iv) $3x + 4y + 5z - 26 = 0$
14. The planes $2x + by + 5z + 12 = 0$ and $3x + 7y + 3z - 4 = 0$ cuts at 90° then $b =$ _____.
(i) $\frac{14}{9}$ (iii) -3
(ii) 3 (iv) $-\frac{14}{9}$
15. The intercepts (a, b, c) made on the coordinate axes (x -axis, y -axis and z -axis respectively) by the plane $x + 2y - 2z = 9$ are _____.
(i) $a = 9, b = 9/2, c = -9/2$ (iii) $a = -9, b = 9/2, c = -9/2$
(ii) $a = 9, b = -9/2, c = 9/2$ (iv) $a = 9, b = 9/2, c = 9/2$
16. If the axes are rectangular, the distance of the origin from the plane $6x - 3y + 2z - 14 = 0$ is _____.
(i) -2 (iii) 2
(ii) $\sqrt{14}$ (iv) $-\sqrt{14}$
17. The angle between the straight line $\frac{x}{2} = \frac{y}{\sqrt{3}} = \frac{z}{\sqrt{2}}$ and the straight line $\frac{x}{\sqrt{3}} = y, z = 0$ is _____.
(i) 60° (iii) 45°
(ii) 30° (iv) 90°
18. The radius of sphere $x^2 + y^2 + z^2 - 2x + 4y + kz = 11$ is 5, then $k =$ _____.
(i) 6 (iii) 4
(ii) -6 (iv) -4
19. If the plane cutting the sphere passes through center of sphere then the corresponding section is called _____.
(i) circle (iii) plane
(ii) small circle (iv) great circle
20. The sum of the three angles of a spherical triangle is _____.
(i) equal to 180° (iii) more than 180°
(ii) less than 180° (iv) more than 540°
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First Semester of BSc Physics Examination March 2018
PD105 Three Dimensional Geometry

Date: 28/03/2018, Wednesday Time: 10:30 a.m. to 01:00 p.m. Maximum Marks: 50

Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION I

Q - II Answer the following questions as directed. (Attempt any 10)

[20]

1. Find the equation to the straight line passing through the point $(3, -4)$ and cutting off intercept, equal but opposite signs from two axes.
2. Find the area of triangle the coordinates of whose angular points respectively $(1, 3)$, $(-7, 6)$ and $(5, -1)$.
3. Find the equation to the straight line, which passes through the point $(1, -2)$ and which is parallel to the straight line $4x + 3y + 5 = 0$.
4. Find the area of quadrilateral the coordinates of whose angular points taken in order are $(1, 1)$, $(6, 0)$, $(6, 2)$, and $(3, 4)$.
5. Find the poalr equation to a straight line.
6. Find the equation to the straight line which passes through the point (x', y') and is perpendicular to the given straight line.
7. Find the coordinates of the points dividing the line segment between the points $(7, 4)$ and $(5, -6)$ in the ratio $2/3$.
8. Find the equations to the tangent to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the straight line $4x + 3y + 5 = 0$.
9. Find the equation to the straight line which passes through the point $(4, 4)$ and are inclined at an angle of 45° to the line $y = 2x$.
10. If the axes are rectangular find the angle between the planes $ax + by + cz + d = 0$ and $a'x + b'y + c'z + d' = 0$.
11. Show that the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar.
12. Show that the line $\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-5}{4}$ is parallel to the plane $4x + 4y - 5z = 0$.

SECTION II

Q - III Answer the following questions as directed.(Attempt Any 5) [30]

1. Let A and B be two points of a great circle on a sphere of center O and let P be the pole of the great circle. Construct and solve the spherical triangle ABP when $AB = 75^\circ$ and $AB = 90^\circ$
2. Solve the spherical triangle given $a = b = 90^\circ$, $c = 60^\circ$. a, b, c are sides of spherical triangle.
3. (i) Find the difference in longitude between, [03]
(a) New York (long. $74^\circ 1.0' W$) and Pearl Harbor (long. $157^\circ 58.3' W$).
(c) New York and Sydney (long. $151^\circ 13.0' E$).
(b) New York and Moscow (long. $37^\circ 34.3' E$).
(d) Sydney and Moscow.
- (ii) Find the distance (in n.m.) between each pair of points on the earth surface. [03]
(a) A (lat. $30^\circ 25' N$, long. $40^\circ W$) and B(lat. $75^\circ 10' N$, long. $40^\circ W$)
(b) A (lat. $30^\circ 25' S$, long. $50^\circ E$) and B(lat. $75^\circ 10' N$, long. $50^\circ E$)
4. For any spherical triangle ABC derive cosine law for sides:

$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

5. Find the equations to the straight lines which pass through a given point (x', y') and make a given angle α with the given straight line $y = mx + c$
 6. Find the area of the triangle, the coordinates of whose angular points are given, the axes being rectangular.
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